

Derivation rules (all three rules apply simultaneously):

$$\begin{array}{c}
 \frac{s \xrightarrow{a} s' \text{ in } TS_1}{(s, t) \xrightarrow{a} (s', t)} \qquad \frac{t \xrightarrow{b} t' \text{ in } TS_2}{(s, t) \xrightarrow{b} (s, t')} \qquad \frac{(a, b) \in H \quad s \xrightarrow{a} s' \text{ in } TS_1 \quad t \xrightarrow{b} t' \text{ in } TS_2}{(s, t) \xrightarrow{(a, b)} (s', t')}
 \end{array}$$

Given: $H = \{(a, b)\}$, and:

TS_1 : states p_0, p_1 , initial p_0 , transitions $p_0 \xrightarrow{a} p_1$ and $p_1 \xrightarrow{c} p_0$.

TS_2 : states q_0, q_1 , initial q_0 , transitions $q_0 \xrightarrow{b} q_1$ and $q_1 \xrightarrow{d} q_0$.

Which transitions leave the initial state (p_0, q_0) in the composition?